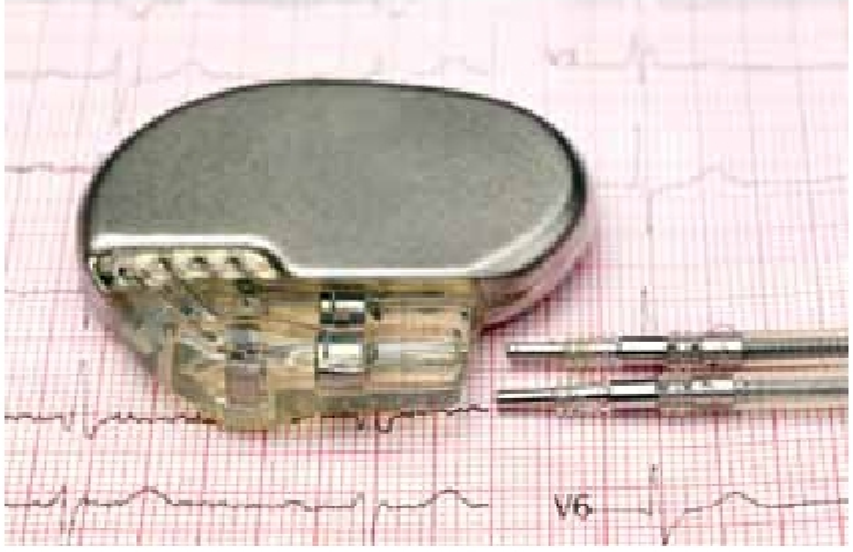

Diamond batteries charge for thousands of years and are apt for devices like pacemakers (Source: www.bhf.org.uk)

Dr Tom Scott—The key person behind the diamond battery innovation (Source: www.bristol.ac.uk)

from decommissioned nuclear power plants. Their work remains confidential, and not much information is available in public. Unlike most of the electricity-generation technologies that use energy to move a magnet through a coil of wire to generate a current, the man-made radioactive diamond produces electric charge simply by placing it near a radioactive source. Carbon-14 or C-14 is created in graphite rods, which are used in power plants to moderate the nuclear reaction.

The radioactive C-14 gets concentrated at the surface of these blocks, from where it is extracted and fused into a diamond to produce a nuclear-powered diamond. This is encapsulated in an artificial diamond to ensure safe containment of the radiation. The lifetime of diamond batteries far exceeds the life of conventional batteries as the half-life of C-14 is as many as 5,730 years. Diamond batteries, which are highly durable, provide charge for thousands of years and serve well as a low-voltage energy source. There are several low-voltage devices like pacemakers, satellites, and space probe components that cannot be easily accessed for replacement of their batteries.

One of the serious problems of using nuclear energy in power plants is the management of nuclear waste that remains radioactive for tens of thousands of years. This exciting technology holds great promise as large amounts of nuclear waste can be converted into a usable electricity source.

How it all started?

Materials engineering professor Dr Tom Scott's research at the University of Bristol is based on understanding the basic and changing properties of materials in various engineered and natural environments to better forecast their performance and create methods for enhancing them. He has spent a considerable amount of time researching for and on behalf of the nuclear industry. He is the technical director of spin-out ImiTec Ltd, a company specialised in the detection and mapping of radioactive materials in industrial, urban, and environmental settings.

In his talk 'Diamonds Are Forever,' the professor opined that we are solving two problems in one shot: the disposal of the build-up of radioactive graphite in spent nuclear fuel (SNF) and the need for a long-lasting source of constant energy.

The first-generation reactors used graphite blocks as moderators to slow down neutrons to keep the nuclear fission process running. Instead of burying radioactive C-14 formed on the surface of graphite, Scott suggested extracting and using it for creating 'deliberately radioactive' diamonds, which when encapsulated inside an artificial diamond could produce a nuclear-powered battery for a long-term supply of clean energy.

He explained that extracting radioactive C-14 can also provide a practical solution to the problem of nuclear waste disposal. This technique was first demonstrated by the researchers in 2016 when they grew a man-made radioactive diamond in the lab, which was placed into a radioactive field to generate a small electric current.

Scientists at the UK Atomic Energy Authority's (UKAEA's) Hydrogen-3 Advanced Technology facility (H3AT) are now working on a pilot project to establish a production line for the batteries capable of the initial supply of up to 20,000 devices per year.

A scientist conducting research on diamond battery technology at H3AT (Source: UKAEA)